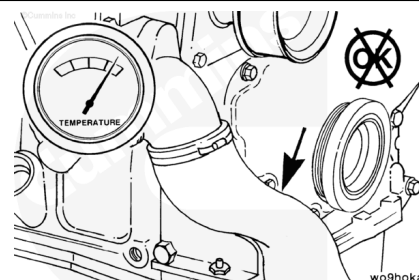


Trainings ▾

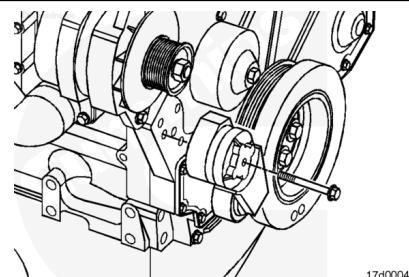
General Information

Note : This procedure is for the engine cooling loop. The engine cooling loop is separate from the low temperature aftercooler (LTA) loop. Refer to Procedure 008-109 (Low-Temperature Aftercooler (LTA) Water Pump) in Section 8 for removal and installation instructions of the LTA pump.
(/qs3/pubsys2/xml/en/procedures/39/39-008-109-tr-gc.html)

The water pump pulls coolant from the bottom of the radiator and pumps it through the engine and back to the top of the radiator for cooling. Reduced or interrupted flow will result in the engine running hot.

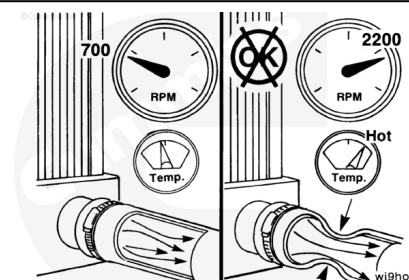


The pump is belt-driven from the crankshaft pulley. An automatic belt tensioner is used to prevent the belt from slipping on the pump pulley. A malfunction of the tensioner will cause the water pump impeller to rotate at a slower speed, reducing the amount of coolant flow.

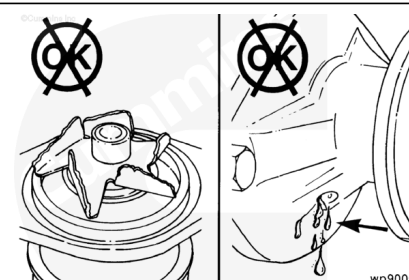


The coolant flow can also be reduced if the inlet hose to the water pump collapses. A hose will usually **not** collapse while the engine is running at low speed. Check the hose while the engine is running at rated speed.

Note : Be sure the engine is warm, a minimum of 91°C [195°F], so the thermostat is open.



A worn or malfunctioning water pump will **not** produce the flow required to prevent the engine from running hot. However, be sure to check the other possibilities indicated in the Coolant Temperature Above Normal symptom chart in Section TS before checking the flow or replacing the pump.

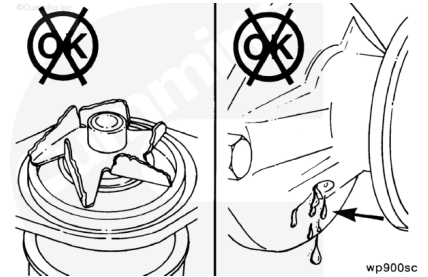


Initial Check

Inspect the water pump housing for cracks and other damage.

Inspect the water pump weep hole for an indication of a steady leak.

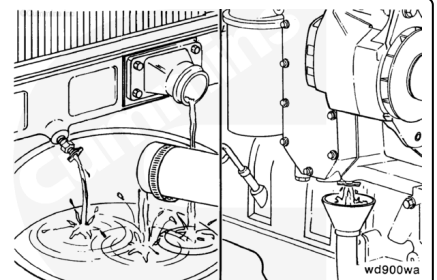
Note : A streak or chemical buildup at the weep hole is not justification for water pump replacement. If a steady flow of coolant is observed, replace the water pump.



Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

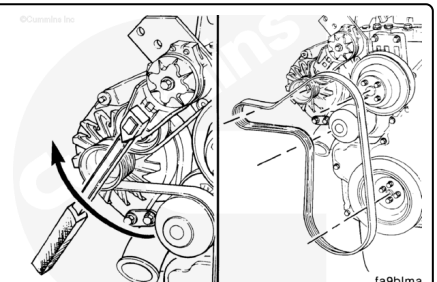


⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

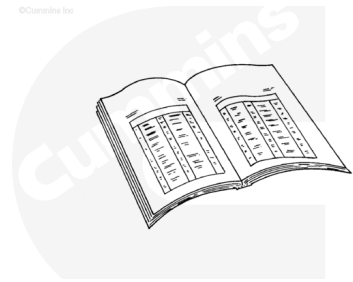
Drain the coolant. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/100/100-008-018.html>).

Remove the drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/100/100-008-002.html>).



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

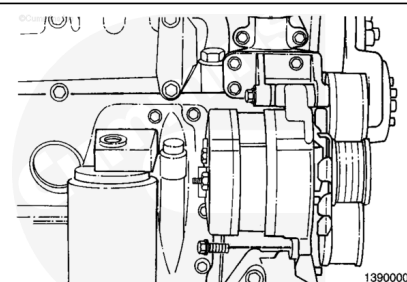
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. Refer to Procedure 013-009 (Battery Cables and Connections) in Section 13.
(/qs3/pubsys2/xml/en/procedures/100/100-013-009-tr-gc.html)
- Drain the engine coolant. Refer to Procedure 008-018 (Cooling System) in Section 8.
(/qs3/pubsys2/xml/en/procedures/100/100-008-018-tr-gc.html)
- Remove the drive belt. Refer to Procedure 008-002 (Drive Belt, Cooling Fan) in Section 8.
(/qs3/pubsys2/xml/en/procedures/100/100-008-002-tr-gc.html)

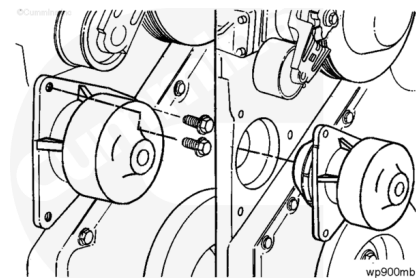
Remove

Remove the alternator link; Refer to Procedure 013-001
(/qs3/pubsys2/xml/en/procedures/100/100-013-001.html).

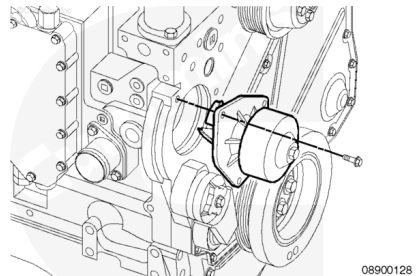


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Remove the water pump mounting capscrews, and water pump.



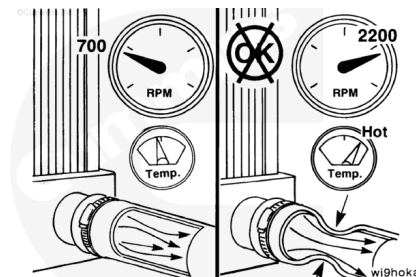
Remove the water pump mounting capscrews and the water pump.



Inspect for Reuse

Inspect the coolant hose. The coolant flow can also be reduced if the inlet hose to the water pump collapses. A hose will usually **not** collapse while the engine is running at low speed. Inspect the hose while the engine is running at rated speed.

Make sure that the engine is warm, so the thermostat is open.

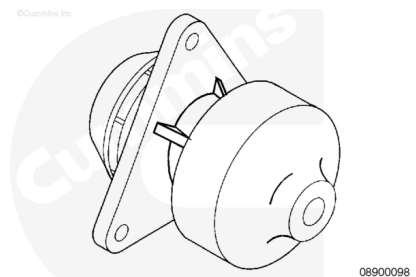


Measurements

	celsius	fahrenheit
Engine Temperature	88	190

The water pump pulls coolant from the bottom of the radiator and pumps it through the engine back to the top of the radiator for cooling.

A worn or malfunctioning water pump will **not** produce the flow required to prevent the engine from running hot. Make sure to inspect the other possibilities indicated in the



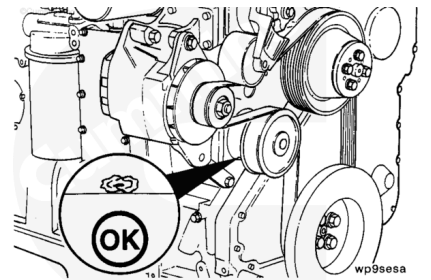
Troubleshooting Logic Chart before inspecting the flow or replacing the water pump.

Check the water pump seal weep hole.

The water pump seal design requires a coolant film for lubrication and cooling. Therefore, it is normal to observe a minor chemical buildup or streaking at the weep hole.

Note : A streak or chemical buildup at the weep hole is not justification for water pump replacement.

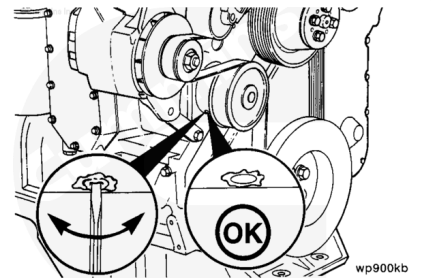
Use the following guidelines to determine if water pump replacement is necessary:



Make sure the weep hole is open.

Note : A small screwdriver or a similar tool can be used to remove any debris.

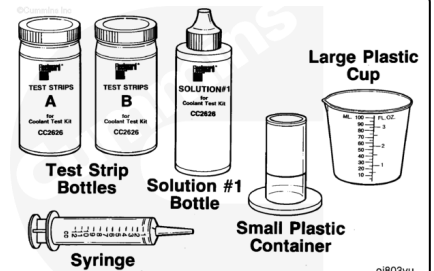
If no leakage is observed from the weep hole under operating conditions, do **not** replace the water pump.



If coolant is seen dripping from the weep hole:

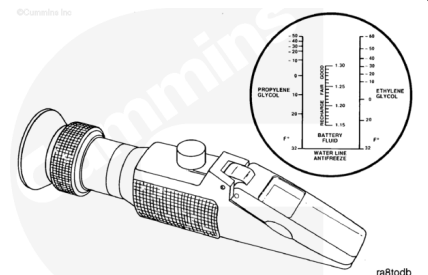
- Use Fleetguard® coolant test kit, Part Number CC-2626, to check the coolant inhibitor level, and adjust the inhibitor level according to the test kit chart.

Note : Supplemental coolant additive concentrations exceeding 2 units per gallon will cause water pump seal leakage.



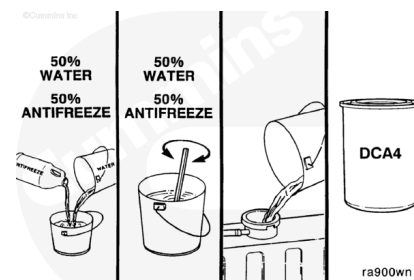
- Use Fleetguard® refractometer, Part Number CC-2800, to check the antifreeze level of the coolant.

Note : Ethylene-glycol-type antifreeze concentration levels exceeding 70 percent will lessen the freezing point protection and will contribute to water pump seal leakage.



If the antifreeze concentration is above 60 percent, drain all the coolant and add makeup water to bring the antifreeze concentration between a minimum of 40 percent to a maximum of 60 percent. Replace the coolant filter and add

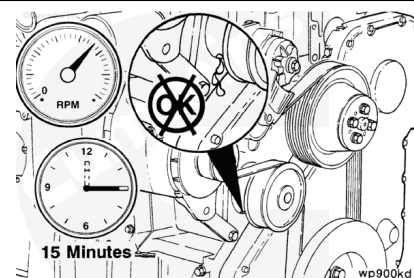
new coolant inhibitor to the recommended levels as outlined in the Operation and Maintenance Manual, C Series Engines, Bulletin Number 3666183.



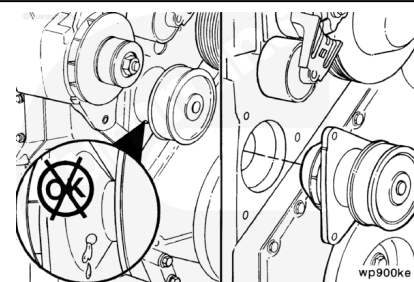
⚠ CAUTION ⚠

If the coolant leakage is observed as a steady stream upon start-up, stop the engine immediately and replace the water pump.

Operate the engine at high idle for 15 minutes with the proper inhibitor and antifreeze levels, and inspect the water pump for leakage.

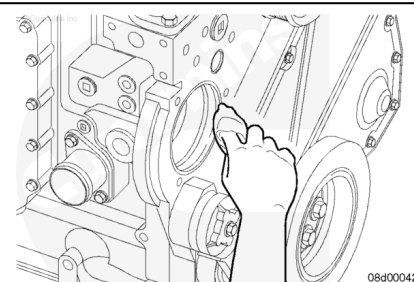


If the coolant is leaking after the above inspections, replace the water pump.



Clean and Inspect for Reuse

Clean the o-ring sealing surface on the water pump housing and the cylinder block.

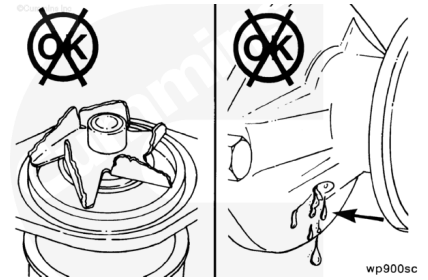


Note : A streak or chemical buildup at the weep hole is not justification for water pump replacement. If a steady flow of coolant or oil is observed, replace the water pump with a new or rebuilt unit.

Inspect the impeller for cracks, missing blades, slippage on the shaft, and other types of damage.

Inspect the water pump housing for cracks and other damage.

Replace the water pump if any damage is found.

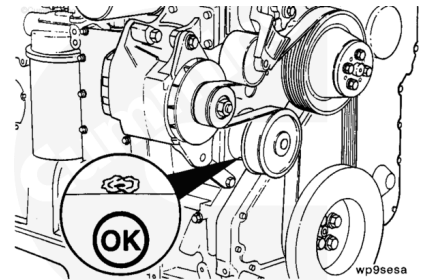


Check the water pump seal weep hole.

The water pump seal design requires a coolant film for lubrication and cooling. Therefore, it is normal to observe a minor chemical buildup or streaking at the weep hole.

Note : A streak or chemical buildup at the weep hole is not justification for water pump replacement.

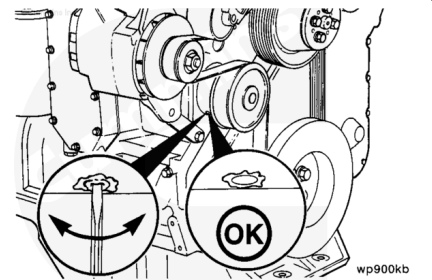
Use the following guidelines to determine if water pump replacement is necessary.



Make sure the weep hole is open.

Note : A small screwdriver or a similar tool can be used to remove any debris.

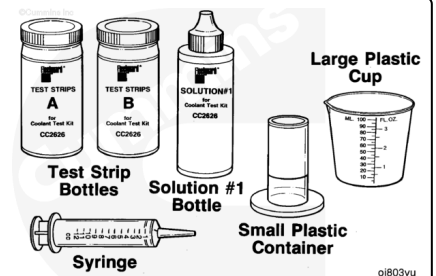
If no leakage is observed from the weep hole under operating conditions, do **not** replace the water pump.



If coolant is seen dripping from the weep hole:

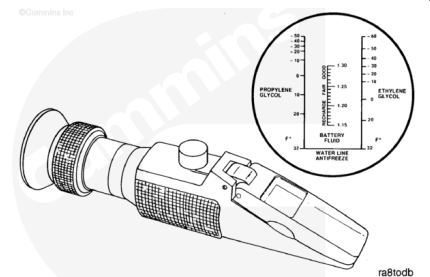
- Use Fleetguard® coolant test kit, Part Number CC-2626, to check the coolant inhibitor level, and adjust the inhibitor level according to the test kit chart.

Supplemental coolant additive concentrations exceeding 2 units per gallon will cause water pump seal leakage.

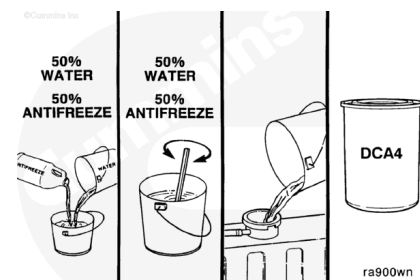


- Use Fleetguard® refractometer, Part Number CC-2800, to check the antifreeze level of the coolant.

Ethylene-glycol-type antifreeze concentration levels exceeding 70 percent will lessen the freezing point protection and will contribute to water pump seal leakage.

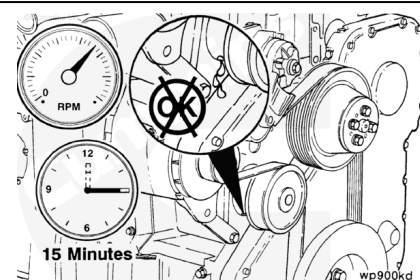


If the antifreeze concentration is above 60 percent, drain all the coolant and add makeup water to bring the antifreeze concentration between a minimum of 40 percent to a maximum of 60 percent. Replace the coolant filter and add new coolant inhibitor to the recommended levels, as outlined in the operation and maintenance manual.



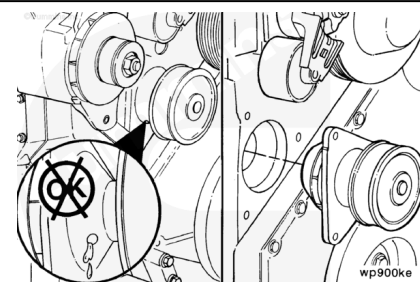
⚠ CAUTION ⚠

If the coolant leakage is observed as a steady stream upon startup, stop the engine immediately and replace the water pump.



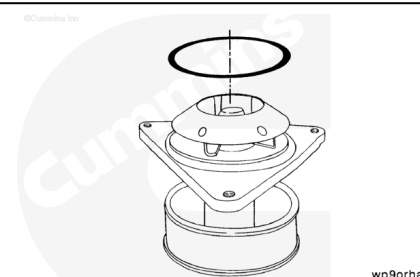
Operate the engine at high idle for 15 minutes with the proper inhibitor and antifreeze levels. Inspect the water pump for leakage.

If the coolant is leaking after the above inspections, replace the water pump.



Install

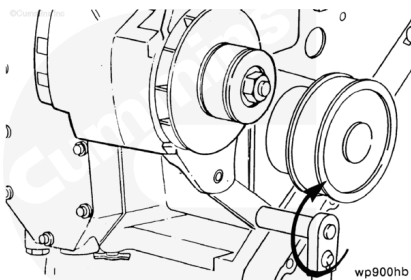
Install a new o-ring into the groove in the water pump.



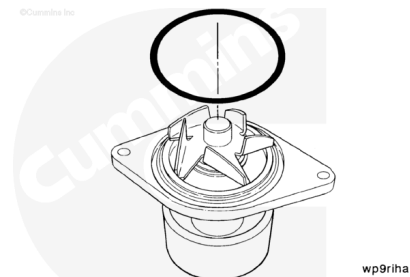
Install the water pump and water pump mounting capscrews.

Tighten the water pump mounting capscrews.

Torque Value: 24 n•m [212 in-lb]

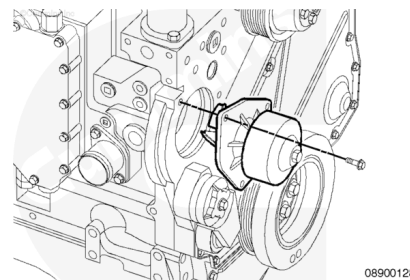


Install a new sealing ring into the groove in the water pump housing.

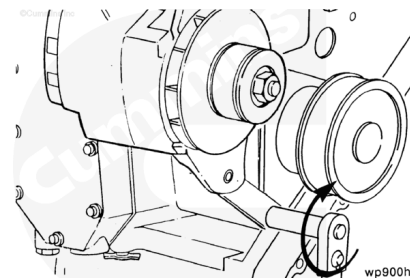


Install the water pump.

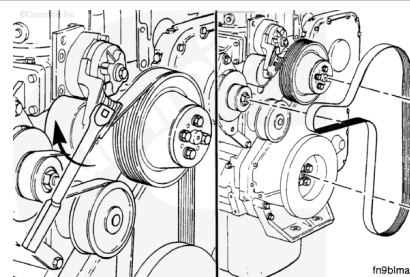
Torque Value: 24 n•m [212 in-lb]



Install and tighten the alternator link; Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/100/100-013-001.html>).



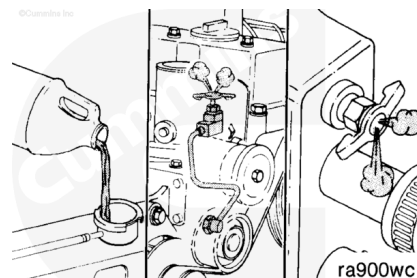
Install the drive belt; Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/100/100-008-002.html>).



During filling, air **must** be vented from the engine coolant passages. Open the engine vent petcock if equipped. Also, be sure to open the petcock on the aftercooler for aftercooled

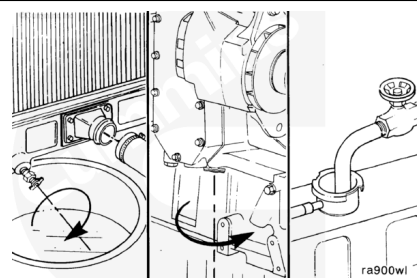
engines.

Note : Venting will permit a fill rate of 19 liters/min. [5 gal/min.].



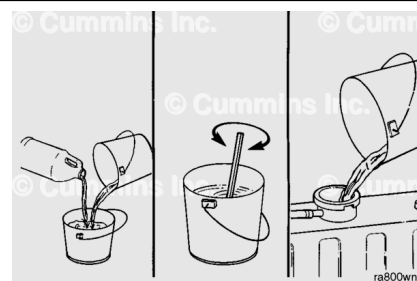
Close the drain valves.

Install all hoses previously removed.



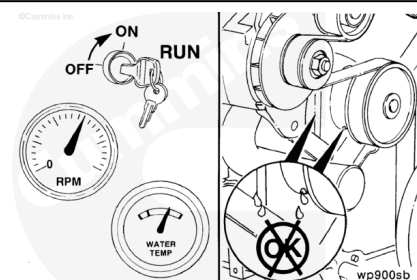
Fill the cooling system with a premixture of 50-percent water and 50-percent ethylene-glycol antifreeze; Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/100/100-008-018.html).

Note : The ability of antifreeze to remove heat from the engine is **not** as good as water; pouring antifreeze into the engine first can contribute to an overheated condition before the liquids are completely mixed.



Install the pressure cap.

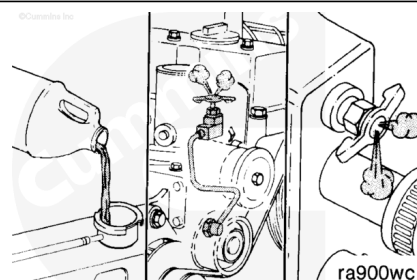
Operate the engine until it reaches a temperature of 80°C [180°F], and inspect for coolant leaks.



Fill

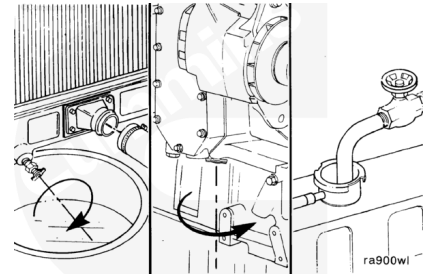
During filling, air **must** be vented from the engine coolant passages. Open the engine vent petcock if equipped. Also, be sure to open the petcock on the aftercooler for aftercooled engines.

Note : Venting will permit a fill rate of 19 liters/min. [5 gal/min.].



Close the drain valves.

Install all hoses previously removed.



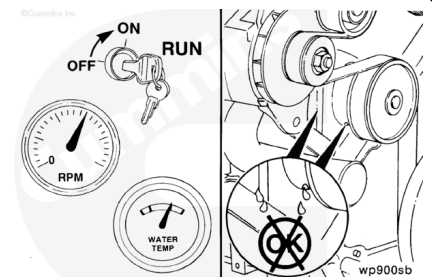
Fill the cooling system with a premixture of 50-percent water and 50-percent ethylene-glycol antifreeze; Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/100/100-008-018.html).

Note : The ability of antifreeze to remove heat from the engine is **not** as good as water; pouring antifreeze into the engine first can contribute to an overheated condition before the liquids are completely mixed.



Install the pressure cap.

Operate the engine until it reaches a temperature of 80°C [176°F], and inspect for coolant leaks.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the drive belt. Refer to Procedure 008-002 (Drive Belt, Cooling Fan) in Section 8. (/qs3/pubsys2/xml/en/procedures/100/100-008-002-tr-gc.html)

- Fill the cooling system. Refer to Procedure 008-018 (Cooling System) in Section 8.
(/qs3/pubsys2/xml/en/procedures/100/100-008-018-tr-gc.html)
- Connect the batteries. Refer to Procedure 013-009 (Battery Cables and Connections) in Section 13.
(/qs3/pubsys2/xml/en/procedures/100/100-013-009-tr-gc.html)
- Operate the engine and check for leaks.

Last Modified: 08-Sep-2016
